

PHASE 3 – PROJECT DESIGN

INTRODUCTION

The Project Design phase defines the structure and working model of the **Import Data using Transform Maps (Spreadsheets)** project. The design explains how raw spreadsheet files, import set staging tables, field maps, and transform maps work together to populate the Institution Details table (u_institution_details).

2.SYSTEM ARCHITECTURE

The proposed system consists of the following components:

- ServiceNow Platform
- Excel / CSV Spreadsheet Source
- Import Set Staging Table (u_institution_details_import)
- Transform Map (Institution Details Transform Map)
- Target Table (u_institution_details)
- Data Validation & Coalesce Rules

Excel / CSV Data File



ServiceNow Import Set

(System Import Sets Module)



Staging Table: u_institution_details_import



Transform Map & Field Mapping Rules



Direct Field

Reference Lookup

Choice Mapping

Mapping

(sys_user)

(ECE/EEE/CSE)



Target Table: u_institution_details

3.TABLE DESIGN & FIELD MAPPING

The data transformation routes fields from the temporary staging table to the custom target table:

Target Table Label: Institution Details

Target Table Name: u_institution_details

Source Field	Target Field	Field Type	Purpose Mapping Rule
u_student_roll_number	student_roll_number	Auto Number	Ignored on insert / Auto-generated by ServiceNow
u_student_name	student_name	Reference – User	Maps to User ID / Name in sys_user table
u_faculty_name	faculty_name	Reference – User	Maps to User ID / Name in sys_user table
u_branch	branch	Choice	Validates choices: ECE, EEE, CSE
u_email	email	String	Stores email details (Set as Coalesce key)
u_phone_number	phone_number	String	Stores email details (Set as Coalesce key)

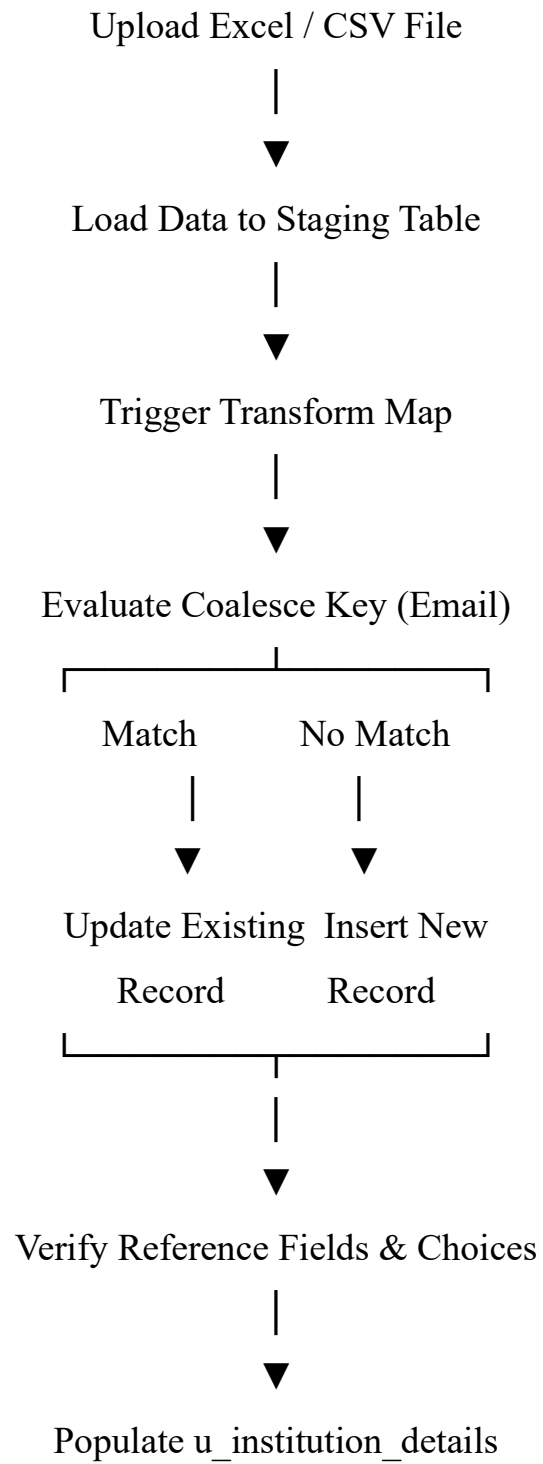
4. TRANSFORM MAP & COALESCE DESIGN

The transformation rules govern how incoming records are processed:

Component	Design Configuration	Purpose
Transform Map Name	Institution Details Transform Map	Core mapping configuration engine
Source Table	u_institution_details_import	Temporary staging table for uploaded raw data
Target Table	u_institution_details	Destination production table
Coalesce Field	email	Prevents duplicate inserts by matching existing records
Choice Action	ignore or create	Handles unmapped branch choices safely

5.DATA TRANSFORM FLOW

The step-by-step processing execution flow:



6. SECURITY & ACCESS DESIGN

- **Data Load Permissions:** Requires admin or import_transformer roles to upload files, create staging tables, and execute transform maps.
- **Post-Import ACL Enforcement:** Once transformed, records residing in u_institution_details are protected by existing Read (bb1), Create (bb2), Write (bb3), and Delete (bb4) ACL policies.

7. TESTING DESIGN

The transformation process will be validated using structured scenarios:

Test Case	Procedure	Expected Result
Standard	Upload Upload sample spreadsheet with ECE, EEE, CSE records	Records correctly loaded into staging table
Field Mapping	Run Transform Map	Target fields populated with correct types and reference links
Coalesce Test	Re-run same dataset with modified description	Existing records updated without creating duplicates
Choice Field Test	Import invalid branch value	System handles choice mapping as designed

8. DESIGN OUTCOME

The Project Design phase establishes:

- The staging and target table schema.
- Source-to-target field mapping parameters.
- Coalesce and duplicate handling strategy.
- The end-to-end data transformation workflow.
- The testing strategy for import validation.

This design serves as the technical blueprint for configuring and executing data imports using Transform Maps in ServiceNow.